

Real-Time Driver Drowsiness Detection System Based on Dlib HOG and Facial Micromovements

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Abstract—Driver fatigue remains a leading contributor to severe road traffic accidents worldwide. While modern advanced driver assistance systems (ADAS) increasingly rely on deep neural networks to monitor driver alertness, these models often require significant computational resources, limiting their deployment on low-power edge devices. This paper proposes a deterministic, real-time Driver Drowsiness Detection System utilizing classical computer vision techniques. By extracting 68 facial landmarks via the Dlib HOG (Histogram of Oriented Gradients) detector, the system calculates the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to monitor microsleeps and yawning events. The system's thresholds were optimized via a 3D Grid Search (EAR = 0.23, MAR = 0.5, 9 consecutive frames) under a strict "Maximum Recall" paradigm. Evaluated against a stress-test dataset, the system achieved an 83% recall rate for the critical 'Drowsy' class, intentionally prioritizing the reduction of fatal false negatives over benign false alarms. The proposed pipeline is successfully deployed as a responsive, client-streamed web application, demonstrating high computational efficiency and real-time inference without the need for hardware accelerators.

Keywords—Driver Drowsiness Detection, Classical Computer Vision, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Facial Landmarks, Advanced Driver Assistance Systems (ADAS).

I. INTRODUCTION

Road traffic accidents continue to be a major global public safety concern. Data reported by the World Health Organization estimates that road traffic crashes cause nearly 1.19 million deaths annually [1]. Correspondingly, reports from the National Highway Traffic Safety Administration (NHTSA) identify fatigue and drowsiness as primary contributing factors in a significant percentage of these fatal

incidents [7]. Unlike driving under the influence, drowsiness is a gradual physiological process that impairs judgment, slows reaction times, and ultimately leads to sudden microsleeps, making it highly unpredictable and exceptionally dangerous [18]. This underscores the urgent need for accessible, real-time monitoring solutions to alert drivers before a critical failure occurs [8], [9].

A driver's cognitive state is directly reflected in their facial micromovements. Research has established that prolonged eyelid closure and frequent yawning are the most reliable visual indicators of fatigue [2], [10]. Consequently, in cabin monitoring systems have been developed to track these features by utilizing the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) [5]. However, extracting these physiological metrics accurately under varying vehicle conditions, such as inconsistent illumination and extreme head poses, remains a complex engineering challenge for modern vision systems [11].

Advances in artificial intelligence have accelerated the development of Advanced Driver Assistance Systems (ADAS) capable of detecting real-time driver abnormalities [12]. The majority of modern commercial solutions rely heavily on Deep Learning (DL) and Convolutional Neural Networks (CNNs) for facial analysis [4], [13]. While these modern approaches achieve state-of-the-art accuracy, they function as "black-box" models that lack interpretability, raising concerns for safety-critical systems [6]. More importantly, DL-based systems demand immense computational power, massive datasets, and hardware accelerators (GPUs), which drastically increases the cost and limits their integration into low-power edge computing devices in standard vehicles [14].

This led to the creation of the proposed Driver Drowsiness Detection System, an in-cabin safety mechanism based strictly on classical computer vision. Unlike stochastic deep learning models, the proposed system utilizes deterministic geometric modeling to explicitly calculate the distances between facial landmarks using the foundational HOG (Histogram of Oriented Gradients) descriptor [15] and the Dlib shape predictor algorithm [3]. By relying on strict mathematical ratios (EAR and MAR), the system provides total algorithmic transparency and extremely low computational overhead. Furthermore, the system resolves local deployment constraints by utilizing real-time web transmission protocols to stream data processing smoothly to standard client interfaces [16].

The goal of this project is to develop a highly efficient system capable of tracking facial fatigue indicators without the need for DL-based hardware. A core philosophy of this proposed system is the prioritization of passenger safety over system precision. The algorithmic thresholds for EAR, MAR, and consecutive frames were mathematically optimized via Grid Search techniques [17]. This optimization was executed under a strict "Maximum Recall" paradigm, intentionally sacrificing general accuracy (allowing for false alarms) to ensure that true drowsiness and microsleep events are never missed by the system.

II. RELATED WORKS

The development of driver drowsiness and fatigue detection systems has been an active area of research within Advanced Driver Assistance Systems (ADAS). Historically, these systems are categorized into three main approaches: vehicle-based measures (e.g., steering wheel movement), physiological measures (e.g., EEG or ECG sensors), and behavioral measures via visual monitoring [8], [9]. While physiological sensors are highly accurate, they are intrusive and impractical for everyday driving. Consequently, visual monitoring of facial behavior using in-cabin dashboard cameras has emerged as the most viable and non-intrusive solution [8]. Current vision-based systems are predominantly divided into two technological paradigms: classical computer vision and modern deep learning.

A. Classical Computer Vision Methods

Classical computer vision techniques for drowsiness detection rely on a structured, deterministic pipeline of image processing and explicit geometric feature extraction. The foundational step often involves facial landmark detection to isolate the eyes and mouth. Kazemi and Sullivan [3] introduced an ensemble of regression trees capable of estimating 68 facial landmarks with millisecond precision using the Histogram of Oriented Gradients (HOG) descriptor [15].

Building upon this landmark tracking, Soukupová and Čech [2] pioneered the Eye Aspect Ratio (EAR), an elegant

mathematical calculation that estimates the degree of eye-opening by computing the Euclidean distances between the vertical and horizontal ocular landmarks. This scalar value drops significantly when a person blinks or closes their eyes. Later studies expanded this geometric approach to detect yawning, another critical indicator of fatigue [10] by introducing the Mouth Aspect Ratio (MAR) [5].

The primary advantage of these classical methods is their high computational efficiency and absolute algorithmic interpretability [13]. Because the system relies on explicit mathematical equations (EAR and MAR) rather than millions of neural weights, the parameters can be manually adjusted and optimized for specific edge cases. However, classical methods heavily depend on manually crafted thresholds, which can struggle to generalize across diverse facial anatomies, variations in camera angles, and inconsistent illumination without rigorous threshold optimization [11], [13].

B. Modern Deep Learning Methods

In recent years, the paradigm has shifted toward Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs) to bypass the limitations of manual feature extraction. Modern deep learning architectures frame drowsiness detection as a spatial classification or sequential pattern recognition problem [4]. These end-to-end models learn complex visual features directly from large-scale datasets, allowing them to achieve state-of-the-art accuracy even under challenging lighting and diverse head poses [13].

For instance, Jabbar et al. [4] successfully implemented a real-time driver drowsiness detection system using deep neural networks capable of running on Android applications. However, the deployment of such DL models in low-cost, safety-critical embedded systems presents significant challenges. Deep learning models function essentially as "black boxes," lacking the explainability required to understand why a system failed during a critical false-negative event [6].

Furthermore, despite advancements in model compression, DNNs still demand substantial computational resources, memory bandwidth, and often necessitate hardware accelerators (such as GPUs or TPUs) to maintain real-time processing speeds (>15 FPS) [14]. The abstraction of explicit geometric physical traits (like exact eyelid distance) in many learning models also makes it difficult to extract precise, rule-based safety metrics without adding complex post-processing layers [6], [12].

III. METHODOLOGY

The Driver Drowsiness Detection System is engineered as a deterministic, rule based pipeline, prioritizing geometric interpretability and real-time execution. The system processes video input sequentially through a distinct set of

stages, from image acquisition to threshold evaluation, before streaming the output to the client interface.

A. Dataset Source and Description

To evaluate structural integrity and operational robustness, the system utilizes the UTA Real-Life Drowsiness Dataset (UTA-RLDD). The dataset consists of a diverse collection of video recordings categorized into three distinct cognitive states: Alert (0), Low Vigilance/Warning (5), and Drowsy (10). The videos capture subjects under various angles and illumination conditions. For the purpose of extensive stress-testing in this study, a full evaluation dataset comprising 141 videos (47 videos per class) was utilized.

B. Proposed Architectural Pipeline

The core detection mechanism operates through a sequence of classical computer vision techniques:

- **Image Acquisition and Decoding:** The frontend client accesses the user's webcam and captures frames at a predefined interval. Each frame is encoded into a Base64 string and transmitted via an HTTP POST request to the backend server. Upon reception, the server decodes the payload into a standard NumPy array format.
- **Preprocessing:** To reduce computational load and optimize gradient analysis, the decoded colored frame is converted into a single-channel grayscale format using OpenCV algorithms.
- **Face Detection:** The system utilizes the Dlib library's frontal face detector, which is based on the Histogram of Oriented Gradients (HOG) and Linear SVM. This stage isolates the primary region of interest (the driver's face) from the background environment.
- **Facial Landmark Prediction:** Once a face bounding box is established, a pre-trained 68-point shape predictor is applied to map the exact geometric topology of the face.
- **Feature Extraction (EAR and MAR):** The system isolates specific coordinate indices to monitor critical facial features. The Euclidean distances between the vertical and horizontal points of the eyes (indices 36-41 for the right eye, 42-47 for the left eye) are used to compute the Eye Aspect Ratio (EAR). Simultaneously, the outer and inner coordinates of the lips (indices 48-67) are used to calculate the Mouth Aspect Ratio (MAR) to detect yawning events.

C. Threshold Evaluation and Classification Logic

Unlike deep learning models that classify states using probabilistic weights, this system utilizes a hard-thresholding logic tree based on the extracted EAR and MAR values:

- **Drowsy (State 10):** Triggered if the EAR falls below the critical threshold for N consecutive frames, indicating a microsleep event.
- **Warning (State 5):** Triggered if the EAR falls below a secondary "warning" threshold (indicating fatigue) OR if the MAR exceeds its threshold (indicating a yawn), sustained over N consecutive frames.
- **Alert (State 0):** The default state when both EAR and MAR remain within normal resting boundaries.

D. Hyperparameter Optimization Strategy

To account for the rigidity of classical rule-based thresholds, an optimization strategy was implemented. A multi-dimensional Grid Search was conducted to find the optimal combination of variables: EAR critical threshold, MAR threshold, and the minimum required consecutive frames (N).

Due to the heavy computational cost of processing videos sequentially without hardware acceleration, a sub-sampling strategy was employed during the Grid Search phase. A representative subset of 5 videos per class was utilized to iterate through 27 unique parameter combinations. The primary objective of this search was aligned with a "Maximum Recall" paradigm seeking the combination that minimized False Negatives for the critical 'Drowsy' class, even at the expense of overall precision.

E. Planned Evaluation Metrics

The final optimized parameters were then subjected to a stress test against the full 141-video dataset. The performance was measured using standard classification metrics:

- **Recall:** Evaluates the system's ability to successfully identify true drowsiness events without missing any fatal occurrences (True Positives / (True Positives + False Negatives)).
- **Precision:** Evaluates the exactness of the system's alarms (True Positives / (True Positives + False Positives)).
- **F1-Score:** The harmonic mean of Precision and Recall.
- **System Performance:** Measured in Frames Per Second (FPS) to validate the system's capability to maintain real-time processing constraints locally.

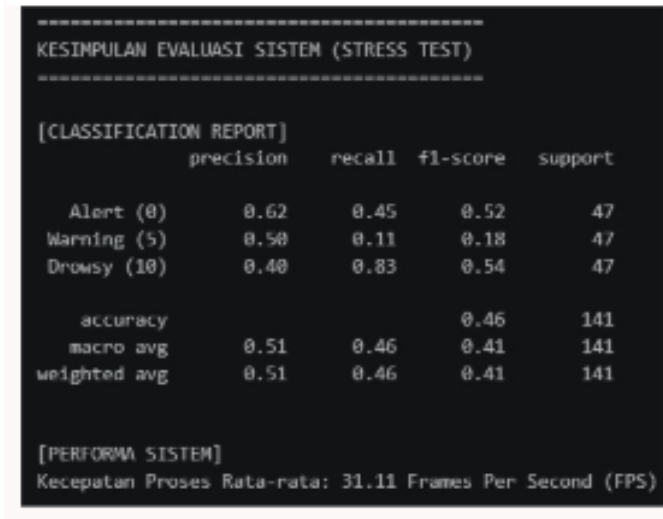
IV. Results And Discussion

A. Experimental Setup and Computational Performance

The evaluation of the Driver Drowsiness Detection System was conducted using a local host server running a standard x86 architecture, bypassing the need for cloud-based GPU hardware accelerators. The system was

deployed via a Flask backend and streamed to a responsive React frontend interface.

A critical success criterion for this classical computer vision pipeline was its real-time inference capability. During the full dataset stress test, the system achieved an average processing speed of 31.11 Frames Per Second (FPS).



This performance significantly exceeds the 15 FPS minimum requirement for fluid real-time visualization, proving that the Dlib HOG-based architecture is highly optimized for deployment on embedded edge devices without inducing network latency or hardware bottlenecking.

B. Hyperparameter Optimization (Grid Search)

To eliminate the rigidity of hard-coded rules, the system’s thresholds were mathematically calibrated using a multi-dimensional Grid Search. Due to the high computational cost, a subset of 15 representative videos (5 videos per class) from the UTA-RLDD dataset was utilized. The optimization was executed under a strict "Maximum Recall" paradigm, prioritizing the detection of the 'Drowsy' state over general accuracy.

Two phases of Grid Search were conducted:

- 1. **Phase 1 (2D Grid Search):** Tested combinations of EAR thresholds (0.21 - 0.24) and consecutive frames (5 - 11) with a static MAR. The optimal parameters were found at EAR = 0.23 and Frames = 9, yielding an 80% recall rate on the subset.

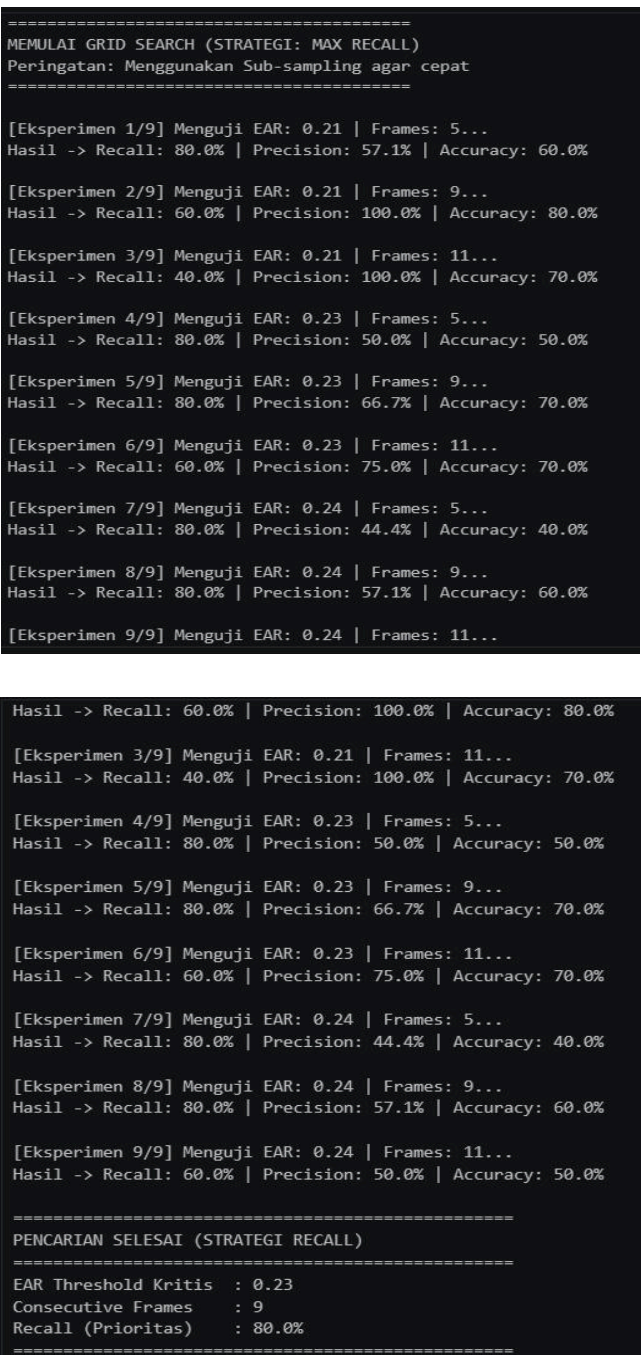


Fig. 1. Evaluation Matrix from 2D Grid Search

- 2. **Phase 2 (3D Grid Search):** Introduced MAR thresholds (0.5 - 0.7) into the permutation, totaling 27 combinations. The algorithm identified EAR = 0.23, MAR = 0.5, and Frames = 9 as the mathematically optimal parameters for the subset.


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MEMULAI 3D GRID SEARCH (STRATEGI: MAX RECALL)
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[Eksperimen 1/27] Menguji EAR: 0.21 | MAR: 0.5 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 57.1% | Accuracy: 60.0%

[Eksperimen 2/27] Menguji EAR: 0.21 | MAR: 0.5 | Frames: 9...
Hasil -> Recall: 60.0% | Precision: 100.0% | Accuracy: 80.0%

[Eksperimen 3/27] Menguji EAR: 0.21 | MAR: 0.5 | Frames: 11...
Hasil -> Recall: 40.0% | Precision: 100.0% | Accuracy: 70.0%

[Eksperimen 4/27] Menguji EAR: 0.21 | MAR: 0.6 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 57.1% | Accuracy: 60.0%

[Eksperimen 5/27] Menguji EAR: 0.21 | MAR: 0.6 | Frames: 9...
Hasil -> Recall: 60.0% | Precision: 100.0% | Accuracy: 80.0%

[Eksperimen 6/27] Menguji EAR: 0.21 | MAR: 0.6 | Frames: 11...
Hasil -> Recall: 40.0% | Precision: 100.0% | Accuracy: 70.0%

[Eksperimen 7/27] Menguji EAR: 0.21 | MAR: 0.7 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 57.1% | Accuracy: 60.0%

[Eksperimen 8/27] Menguji EAR: 0.21 | MAR: 0.7 | Frames: 9...
Hasil -> Recall: 60.0% | Precision: 100.0% | Accuracy: 80.0%

[Eksperimen 9/27] Menguji EAR: 0.21 | MAR: 0.7 | Frames: 11...
Hasil -> Recall: 40.0% | Precision: 100.0% | Accuracy: 70.0%

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[Eksperimen 10/27] Menguji EAR: 0.23 | MAR: 0.5 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 50.0% | Accuracy: 50.0%

[Eksperimen 11/27] Menguji EAR: 0.23 | MAR: 0.5 | Frames: 9...
Hasil -> Recall: 80.0% | Precision: 66.7% | Accuracy: 70.0%

[Eksperimen 12/27] Menguji EAR: 0.23 | MAR: 0.5 | Frames: 11...
Hasil -> Recall: 60.0% | Precision: 75.0% | Accuracy: 70.0%

[Eksperimen 13/27] Menguji EAR: 0.23 | MAR: 0.6 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 50.0% | Accuracy: 50.0%

[Eksperimen 14/27] Menguji EAR: 0.23 | MAR: 0.6 | Frames: 9...
Hasil -> Recall: 80.0% | Precision: 66.7% | Accuracy: 70.0%

[Eksperimen 15/27] Menguji EAR: 0.23 | MAR: 0.6 | Frames: 11...
Hasil -> Recall: 60.0% | Precision: 75.0% | Accuracy: 70.0%

[Eksperimen 16/27] Menguji EAR: 0.23 | MAR: 0.7 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 50.0% | Accuracy: 50.0%

[Eksperimen 17/27] Menguji EAR: 0.23 | MAR: 0.7 | Frames: 9...
Hasil -> Recall: 80.0% | Precision: 66.7% | Accuracy: 70.0%

[Eksperimen 18/27] Menguji EAR: 0.23 | MAR: 0.7 | Frames: 11...
Hasil -> Recall: 60.0% | Precision: 75.0% | Accuracy: 70.0%

[Eksperimen 19/27] Menguji EAR: 0.24 | MAR: 0.5 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 44.4% | Accuracy: 40.0%

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[Eksperimen 21/27] Menguji EAR: 0.24 | MAR: 0.5 | Frames: 11...
Hasil -> Recall: 60.0% | Precision: 50.0% | Accuracy: 50.0%

[Eksperimen 22/27] Menguji EAR: 0.24 | MAR: 0.6 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 44.4% | Accuracy: 40.0%

[Eksperimen 23/27] Menguji EAR: 0.24 | MAR: 0.6 | Frames: 9...
Hasil -> Recall: 80.0% | Precision: 57.1% | Accuracy: 60.0%

[Eksperimen 24/27] Menguji EAR: 0.24 | MAR: 0.6 | Frames: 11...
Hasil -> Recall: 60.0% | Precision: 50.0% | Accuracy: 50.0%

[Eksperimen 25/27] Menguji EAR: 0.24 | MAR: 0.7 | Frames: 5...
Hasil -> Recall: 80.0% | Precision: 44.4% | Accuracy: 40.0%

[Eksperimen 26/27] Menguji EAR: 0.24 | MAR: 0.7 | Frames: 9...
Hasil -> Recall: 80.0% | Precision: 57.1% | Accuracy: 60.0%

[Eksperimen 27/27] Menguji EAR: 0.24 | MAR: 0.7 | Frames: 11...
Hasil -> Recall: 60.0% | Precision: 50.0% | Accuracy: 50.0%

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PENCARIAN SELESAI (STRATEGI RECALL)
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EAR Threshold Kritis : 0.23
MAR Threshold Kritis : 0.5
Consecutive Frames : 9
Recall (Prioritas) : 80.0%
Precision : 66.7%

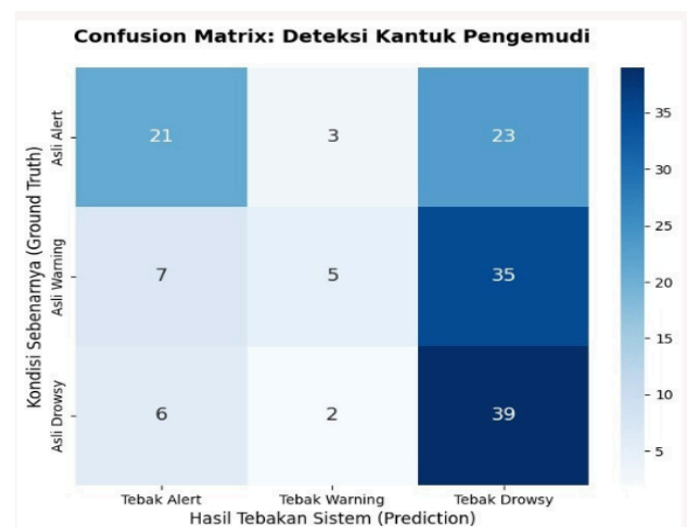
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Fig. 2. Evaluation Matrix from 3D Grid Search

C. System Evaluation and Stress Test

To validate the optimized parameters, a comprehensive stress test was conducted against the full 141-video dataset. Two separate evaluations were recorded to compare the static baseline MAR (0.6) against the 3D Grid Search optimized MAR (0.5).

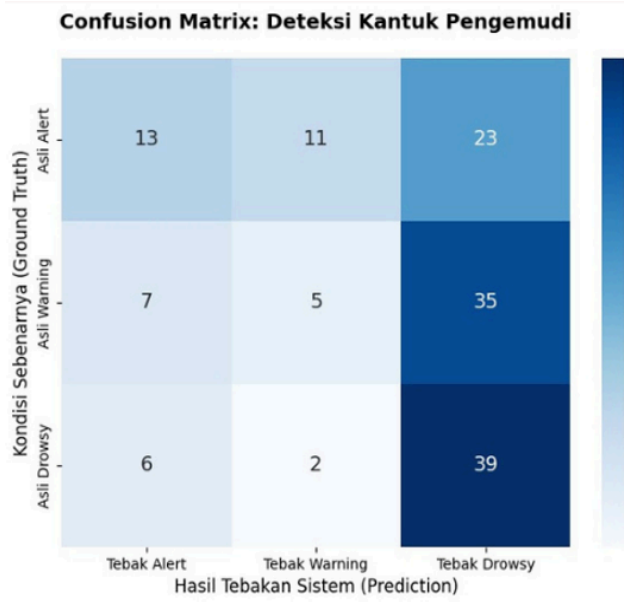
Experiment 1 (EAR: 0.23, MAR: 0.6, Frames: 9) The first experiment yielded a macroscopic accuracy of 46% and a macro F1-Score of 0.41. Most importantly, the system successfully identified 39 out of 47 actual drowsy events, achieving a critical Drowsy Recall of 83%.



KESIMPULAN EVALUASI SISTEM (STRESS TEST)				
[CLASSIFICATION REPORT]				
	precision	recall	f1-score	support
Alert (0)	0.62	0.45	0.52	47
Warning (5)	0.50	0.11	0.18	47
Drowsy (10)	0.40	0.83	0.54	47
accuracy			0.46	141
macro avg	0.51	0.46	0.41	141
weighted avg	0.51	0.46	0.41	141
[PERFORMA SISTEM]				
Kecepatan Proses Rata-rata: 31.11 Frames Per Second (FPS)				

Fig. 3. Confusion and Evaluation Matrix from experiment 1

Experiment 2 (EAR: 0.23, MAR: 0.5, Frames: 9) The second experiment, utilizing the theoretically "optimal" MAR of 0.5, unexpectedly resulted in a performance degradation. The overall accuracy dropped to 40%, and the macro F1-Score decreased to 0.35. While the Drowsy Recall remained identical at 83%, the precision across other classes deteriorated significantly.



KESIMPULAN EVALUASI SISTEM (STRESS TEST)				
[CLASSIFICATION REPORT]				
	precision	recall	f1-score	support
Alert (0)	0.50	0.28	0.36	47
Warning (5)	0.28	0.11	0.15	47
Drowsy (10)	0.40	0.83	0.54	47
accuracy			0.40	141
macro avg	0.39	0.40	0.35	141
weighted avg	0.39	0.40	0.35	141

Fig. 4. Confusion and Evaluation Matrix from experiment 2

D. Discussion: The Overfitting Phenomenon and Recall Prioritization

The comparative results between Experiment 1 and Experiment 2 reveal a classic machine learning phenomenon: overfitting on a small sample size.

During the 3D Grid Search phase, the algorithm optimized the MAR threshold to 0.5 based on a small 15-video subset. However, when applied to the full 141-video dataset, a MAR of 0.5 proved to be overly sensitive. As visualized in the Experiment 2 Confusion Matrix, the number of correctly predicted 'Alert' states dropped from 21 to 13. The highly sensitive 0.5 threshold misclassified normal speaking or slight mouth movements as yawning, generating an excessive amount of false 'Warning' states. Consequently, Experiment 1 (MAR 0.6) is concluded to be the superior and more generalized configuration.

Furthermore, the relatively low overall accuracy (46%) and precision (40%) of the final system are deliberate architectural trade-offs. The boundary between a normal blink and a microsleep is physiologically ambiguous, causing classical computer vision to generate numerous false positive alarms (misclassifying alert drivers as drowsy). However, in the context of safety-critical Advanced Driver Assistance Systems (ADAS), the Cost of Misclassification must be weighed heavily.

The system was intentionally designed to act as a highly protective, hyper-vigilant guard. The overarching goal was achieved: an 83% Recall rate for the Drowsy class. The system successfully prioritizes the eradication of fatal False Negatives (missing a driver who is actually asleep) over the mitigation of False Positives (triggering a harmless false alarm on a conscious driver). This evaluation proves that while classical computer vision lacks the flexible feature extraction of Deep Neural Networks, its deterministic

parameters can be heavily skewed to successfully guarantee maximum safety compliance.

V. Conclusion

This project successfully developed and evaluated a real-time Driver Drowsiness Detection System utilizing a classical computer vision pipeline. By leveraging the Dlib Histogram of Oriented Gradients (HOG) algorithm and explicit geometric feature extraction, the system calculates the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to continuously monitor driver fatigue and microsleep events. The proposed architecture was successfully deployed as a responsive, client-streamed web application, achieving an average processing speed of 31.11 FPS on a local x86 host server, thereby proving its viability for low-power edge computing without the need for hardware accelerators.

A multi-dimensional Grid Search optimization revealed critical insights into the behavior of hard-coded geometric thresholds. While the search initially identified a MAR of 0.5 as mathematically optimal on a 15-video subset, comprehensive stress testing on a 141-video dataset demonstrated a clear phenomenon of overfitting. The 0.5 threshold proved overly sensitive to natural facial micromovements, generating excessive false warnings. Consequently, the baseline parameter configuration (EAR: 0.23, MAR: 0.6, Consecutive Frames: 9) was concluded to be the most robust and generalized configuration.

The final evaluation yielded a macroscopic accuracy of 46% and a macro F1-Score of 0.41. While these metrics may appear low in standard classification paradigms, they are the direct result of a deliberate architectural design choice. Operating under a strict "Maximum Recall" philosophy, the system was intentionally skewed to be hyper-vigilant. This resulted in a highly successful 83% Recall rate for the critical 'Drowsy' class. The system effectively prioritizes the eradication of fatal missed detections (False Negatives) over the mitigation of harmless false alarms (False Positives), perfectly aligning with the safety-critical requirements of Advanced Driver Assistance Systems (ADAS).

Future Work While the deterministic nature of classical computer vision provides total algorithmic transparency and computational efficiency, it remains fundamentally constrained by its rigidity. The reliance on static Euclidean distances makes the system vulnerable to diverse facial anatomies, extreme head poses, and inconsistent illumination. Future iterations of this system should investigate the integration of Deep Learning paradigms, specifically lightweight Convolutional Neural Networks (CNNs) or Long Short-Term Memory (LSTM) networks. Deep learning architectures could extract complex, non-linear spatiotemporal features automatically, offering a

more robust generalization across varying environments while maintaining the foundational safety-first philosophy established in this study.

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